MH1812
Extra Exercises of Recurrence Relations

These are additional recurrence relation questions and may not be discussed in the tutorials.

Q1: Let a_n denote the number of ternary strings (each character of the string is 0, 1 or 2) of length $n \geq 1$ that do not contain consecutive symbols that are the same. Find a recurrence relation for a_n and deduce an explicit formula for a_n.

Solution: For each valid string of length $n-1$, there are always 2 possible characters to append it to form a valid string of length n. All valid strings of length n can be formed in this way. Hence $a_n = 2a_{n-1}$. With the initial value $a_1 = 3$, we have $a_n = 3 \cdot 2^{n-1}$. $\square$

Q2: (a) A pair of newborn rabbits (one of each sex) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. Find a recurrence relation for the number a_n of pairs of rabbits on the island after n months, assuming that no rabbits ever die.

(b) Suppose that in part (a), a pair of rabbits leaves the island after reproducing twice. Find a recurrence relation for the number b_n of rabbits on the island in the middle of the n-th month.

Solution:

(a) $a_n = a_{n-1} + a_{n-2}$, where a_{n-1} comes from the fact that all rabbits from the previous months continue to be alive and a_{n-2} is the number of new pairs of rabbits.

(b) Let $b_{n,i}$ denote the number of pairs of rabbits that are i months old at the beginning of the n-th month. Then

$$b_n = b_{n,0} + b_{n,1} + b_{n,2} \tag{1}$$

$$b_{n,0} = b_{n-1,1} + b_{n-1,2} \tag{2}$$

$$b_{n,i} = b_{n-1,i-1}, \quad i = 1, 2 \tag{3}$$

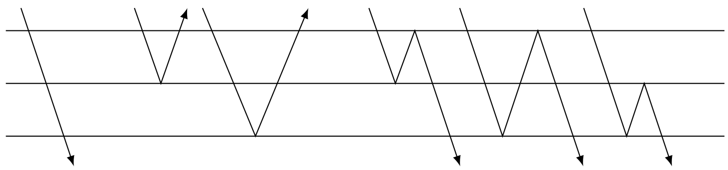
It follows that

$$\begin{aligned}
b_n &= b_{n,0} + b_{n,1} + b_{n,2} \\
&= (b_{n-1,1} + b_{n-1,2}) + b_{n-1,0} + b_{n-1,1} \\
&= (b_{n-2,0} + b_{n-2,1}) + (b_{n-2,1} + b_{n-2,2}) + b_{n-2,0} \\
&= (b_{n-2,0} + b_{n-2,1}) + (b_{n-3,0} + b_{n-2,2}) + (b_{n-3,1} + b_{n-3,2}) \\
&= (b_{n-2,0} + b_{n-2,1} + b_{n-2,2}) + (b_{n-3,0} + b_{n-3,1} + b_{n-3,2}) \\
&= b_{n-2} + b_{n-3}.
\end{aligned}$$

$\square$

Q3: Suppose that we put two panes of glass back-to-back. How many ways a_n are there for light rays to pass through or be reflected after changing direction n times? Find a recurrence relation for a_n. Compare it with the numbers a_n in Q2(a). What do you find?

The first few cases of a_n are:

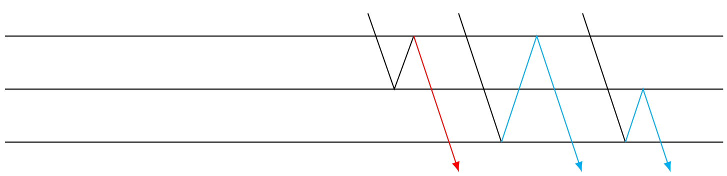


$a_0 = 1$ $\qquad$ $a_1 = 2$ $\qquad$ $a_2 = 3$

Solution: The recurrence relation is $a_n = a_{n-1} + a_{n-2}$, the same as in Q3(a), but the initial values are shifted. Hence, if we let a'_n to denote the numbers in Q3(a), we have $a_n = a'_{n+2}$.

Next we show the correctness of the recurrence relation. When $n \geq 2$, a ray either takes its first bounce off the opposite surface and then continue in a_{n-1} ways, or it begins by bouncing off the middle surface and then bouncing back again to finish in a_{n-2} ways. This shows that $a_n = a_{n-1} + a_{n-2}$.

(Example: The following figure shows the composition of a_2. The red part corresponds to a_0 and the blue part a_1.)



$\square$

Q4: The numbers $a_1, a_2, a_3, \ldots$ defined in Q2(a) is called **Fibonacci numbers**, where a_n is referred to as the n-th Fibonacci number.

(a) Show that a_n equals the integer which is closest to the number

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

(b) Find a general formula for $a_1 + a_3 + \cdots + a_{2n-1}$. (Evaluate the sum for a few small values of n, conjecture a general formula and then prove it.)

(c) Show that a_n is divisible by a_3 if and only if n is divisible by 3.

Solution:

(a) The recurrence relation is $a_n = a_{n-1} + a_{n-2}$ with $a_1 = a_2 = 1$. Solving the recurrence relation, we obtain that

$$a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n =: A_n - B_n \tag{4}$$

To show that a_n is the integer which is closest to A_n, it suffices to show that $|B_n| < 1/2$. Indeed,

$$|B_n| = \frac{1}{\sqrt{5}} \left| \frac{1-\sqrt{5}}{2} \right|^n \leq \frac{1}{\sqrt{5}} \left| \frac{1-\sqrt{5}}{2} \right| = \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}-1}{2} < \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{2} = \frac{1}{2}.$$

(b) Let us first calculuate the sum for the first few small values of n:

$$\begin{aligned} a_1 &= 1 = a_2 \\ a_1 + a_3 &= 1 + 2 = 3 = a_4 \\ a_1 + a_3 + a_5 &= 3 + 5 = 8 = a_6 \\ a_1 + a_3 + a_5 + a_7 &= 8 + 13 = 21 = a_8 \end{aligned}$$

We shall prove by mathematical induction that

$$a_1 + a_3 + \cdots + a_{2n-1} = a_{2n}. \tag{5}$$

The base step is $n = 1$: we have verified that $a_1 = 1 = a_2$, so the claim is true.

Inductive step: Suppose that (5) is true for $n = k$, we shall show that it is true for $n = k + 1$, that is, we need to show that

$$a_1 + a_3 + \cdots + a_{2k-1} + a_{2k+1} = a_{2k+2}.$$

By the induction hypothesis, the left-hand side is $a_{2k} + a_{2k+1} = a_{2k+2}$, which is exactly the right-hand side. Hence, the proof of the inductive step is complete.

Therefore, by mathematical induction, (5) holds for all integers $n \geq 1$.

(c) The statement can be equivalently rephrased as: a_n is even if and only if n is divisible by 3. Noting that $a_1 = a_2 = 1$ are odd and $a_3 = 2$ is even, it suffices to prove that $a_{n+3} - a_n$ is even. Indeed,

$$a_{n+3} - a_n = (a_{n+2} + a_{n+1}) - a_n = (a_{n+1} + a_n) + a_{n+1} - a_n = 2a_{n+1}$$

is even.

Alternatively, one can use the strong induction to show that $a_{n+3} - a_n$ is even. The details are omitted. □